CPE 434

VLSI Design

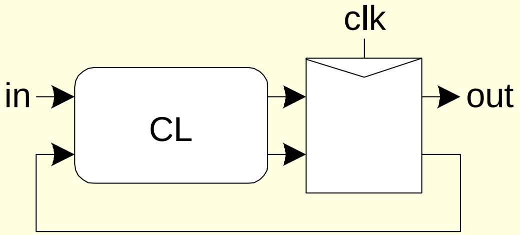
Chapter 10:
Sequential Circuit Design

Lecture Content

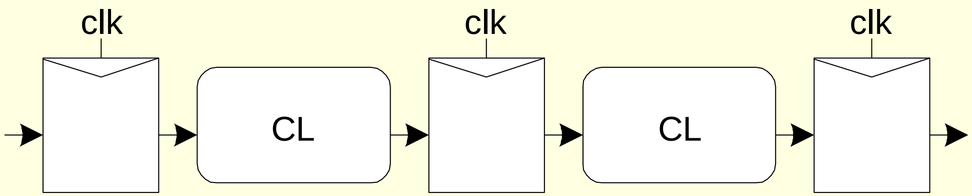
- Sequencing
- Sequencing Element Design
  - D Latch
  - D Flip-Flop
- Sequencing Methods
- Timing Diagrams
- Max-Delay
- Min-Delay

Sequencing

- ***Combinational logic***
  - Output depends on current inputs
- ***Sequential logic***
  - Output depends on current and previous inputs
  - Such circuits are said to have *state*
  - Designed with flip-flops or Latches called memory elements that hold data called *state* or *tokens*
  - Purpose is not really memory, instead, to enforce sequence, therefore, called sequencing elements
  - Distinguishing/separating the current *tokens* from previous & future
  - Ex: FSM, pipeline



Finite State Machine



Pipeline

Sequencing Overhead

- Delay fast tokens so they don’t catch slow ones.
- Use flip-flops to delay fast tokens so they move through exactly one stage each cycle.
- Inevitably adds some delay to the slow tokens
- Makes circuit slower than just the logic delay
  - Called sequencing overhead
- Some people call this clocking overhead
  - But it applies to asynchronous circuits too
  - Inevitable side effect of maintaining sequence

Sequencing Elements

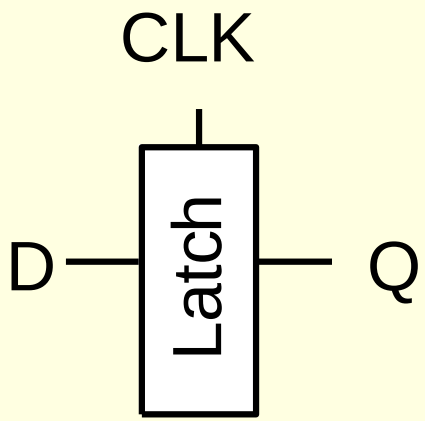
- Either static or dynamic.
- With static storage employs some sort of feedback to retain its output value indefinitely.
- With dynamic storage generally maintains its value as charge on a capacitor that will leak away if not refreshed for a long period of time.

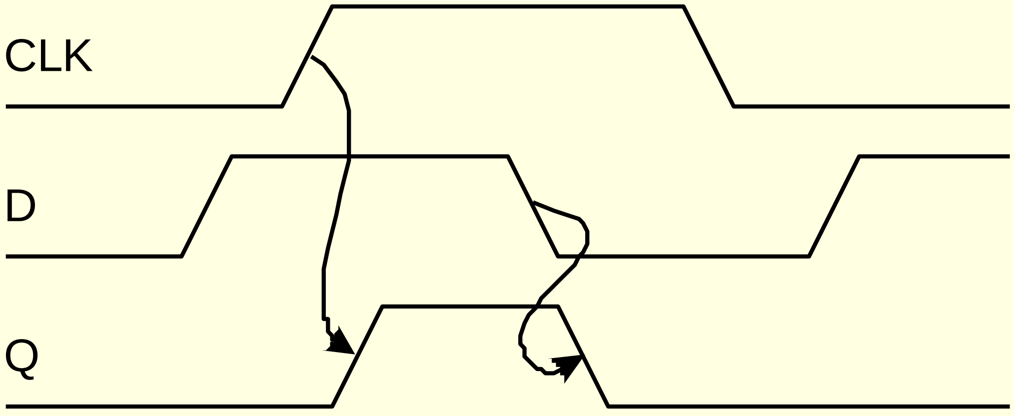
Sequencing Elements

- **Latch**: Level sensitive
  - a.k.a. transparent latch, D latch
- **Flip-flop**: edge triggered
  - A.k.a. master-slave flip-flop, D flip-flop, D register

D Latch

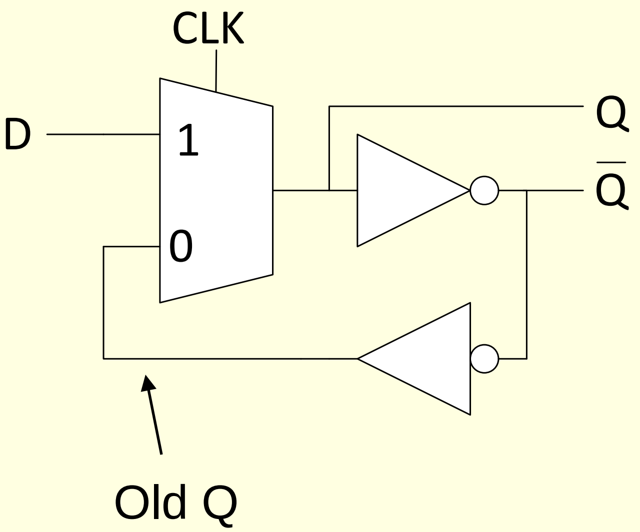
- When CLK = 1, latch is *transparent*
  - Q follows D (a buffer with a Delay)
- When CLK = 0, the latch is *opaque*
  - Q holds its last value independent of D
- a.k.a. *transparent latch* or *level-sensitive latch*

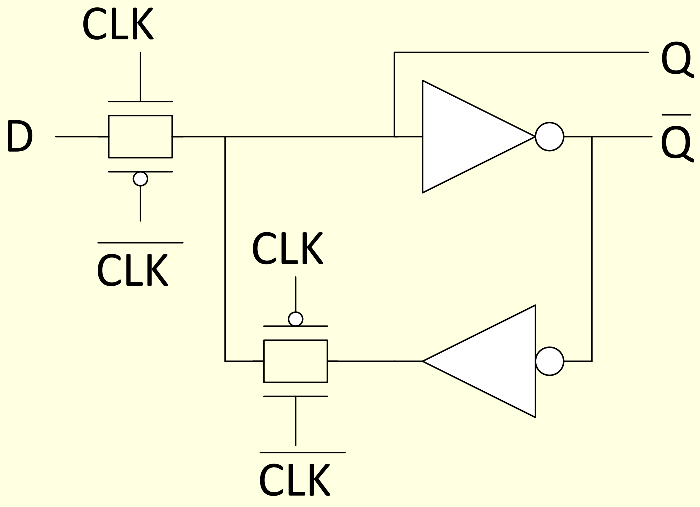




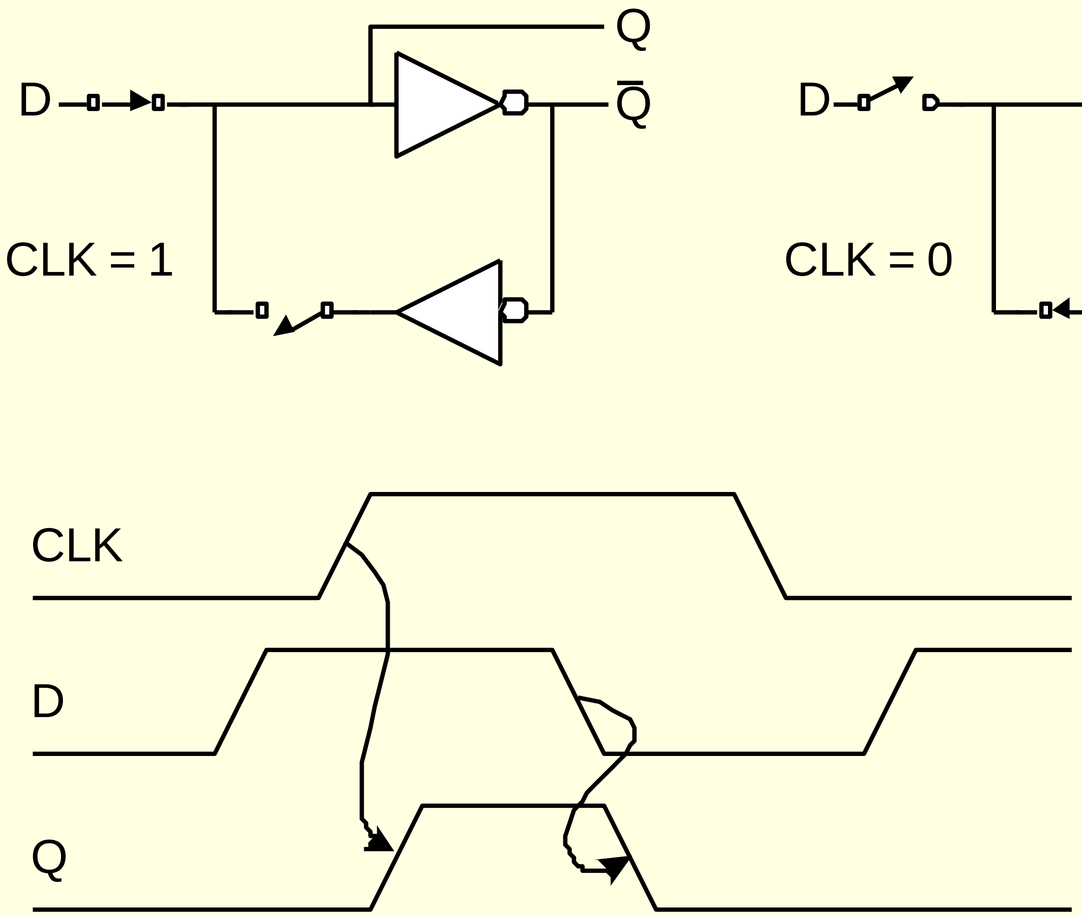
D Latch Design

- Multiplexer chooses D or old Q



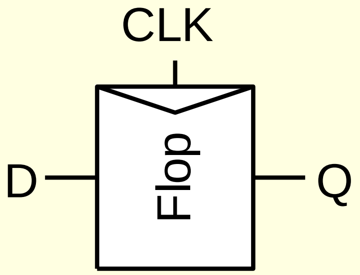


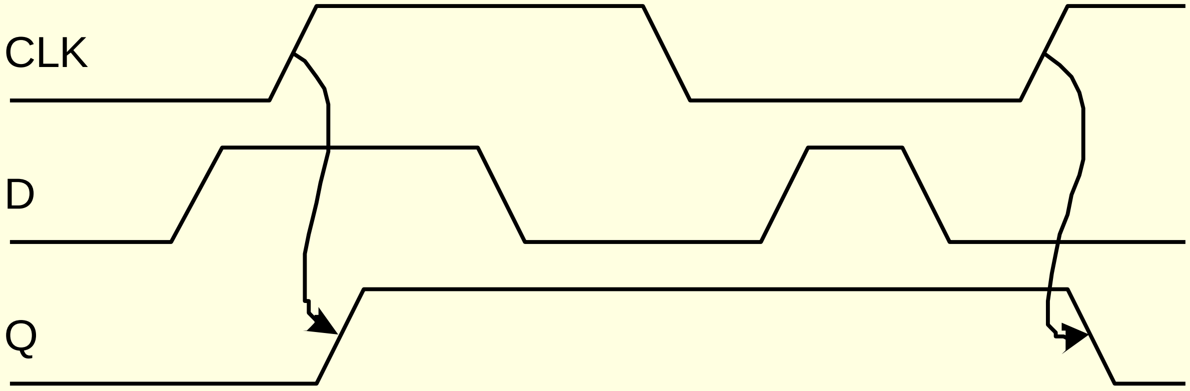
D Latch Operation



D Flip-flop

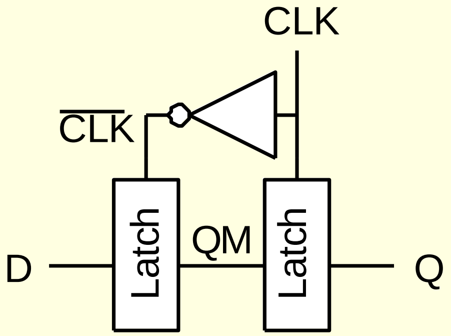
- When CLK rises, D is copied to Q
- At all other times, Q holds its value
- a.k.a. *positive edge-triggered flip-flop*, *master-slave flip-flop*

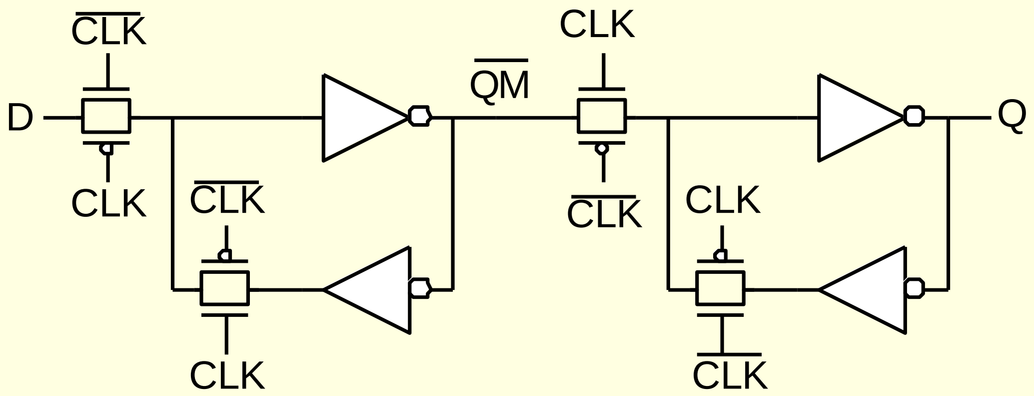




D Flip-flop Design

- Built from master and slave D latches

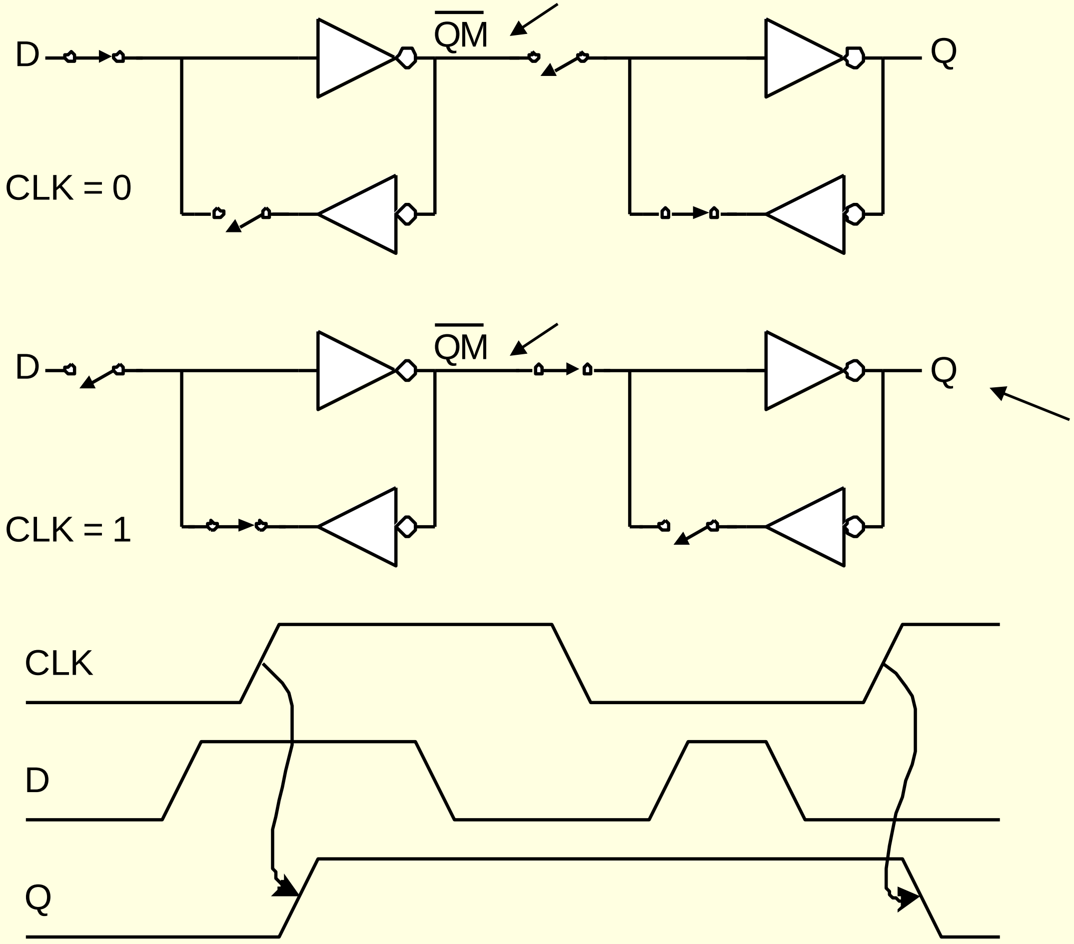




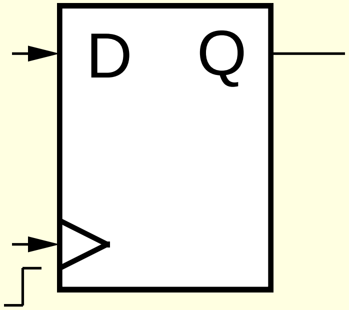
A “negative level-sensitive” latch

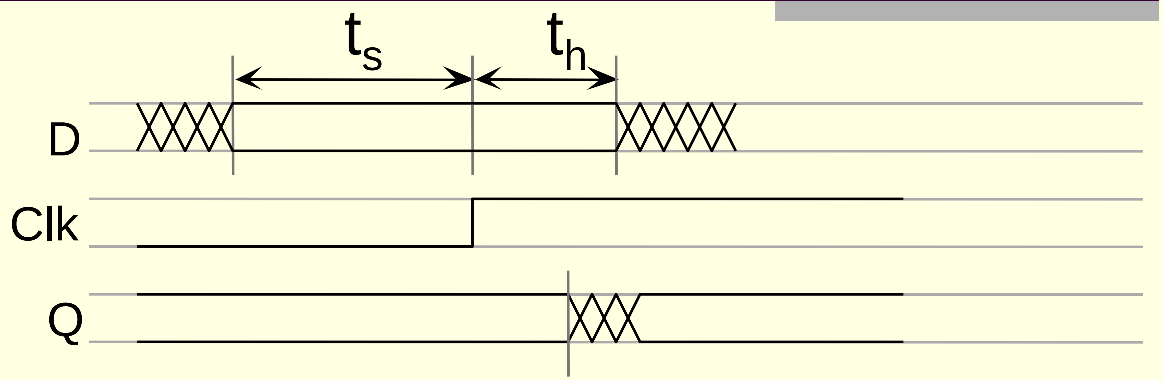
A “positive level-sensitive” latch

D Flip-flop Operation



Definitions





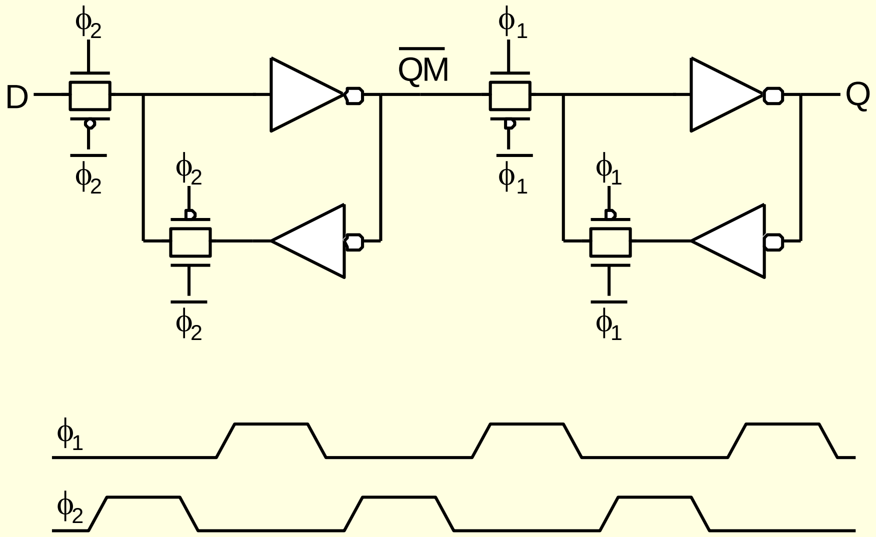
- Timing parameters for clocked devices are specified in relation to the clock input (rising edge)
- D input must be valid at least t_s (setup time) before the rising clock edge
- D input must be held steady t_h (hold time) after rising clock edge
- Setup and hold are input restrictions
  - Failure to meet restrictions causes circuit to operate incorrectly

Race Condition

- Back-to-back flops can malfunction from clock skew because of variations in clock arrival times:
  - Second flip-flop fires Early
  - Sees first flip-flop change and captures its result
  - Called *hold-time failure* or *race condition*

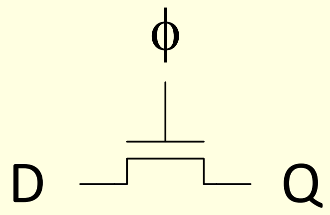
Nonoverlapping Clocks

- Nonoverlapping clocks can prevent races
  - As long as nonoverlap exceeds clock skew
- Good for safe design
  - Industry manages skew more carefully instead



Latch Design

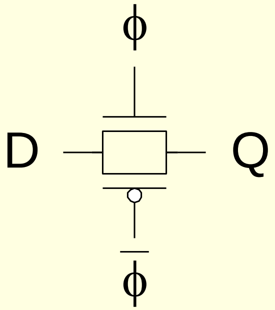
- Pass Transistor Latch
- Pros
  - + Tiny
  - + Low clock load
- Cons
  - – V_t drop
  - – Nonrestoring
  - – Backdriving
  - – Output noise sensitivity
  - – Dynamic output
  - – Diffusion input



Used in 1970’s

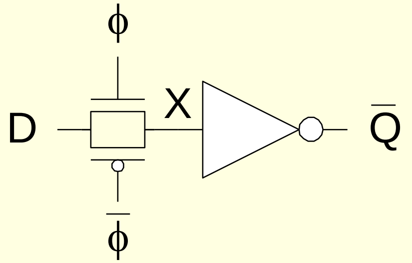
Latch Design

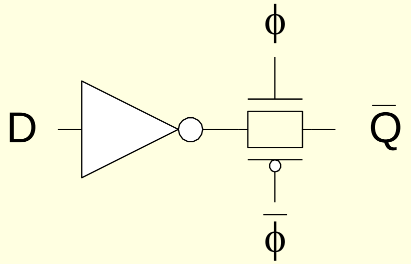
- Transmission gate
- Pros
  - + No V_t drop
- Cons
  - – Requires inverted clock
  - – Backdriving
  - – Output noise sensitivity
  - – Dynamic
  - – Diffusion input



Latch Design

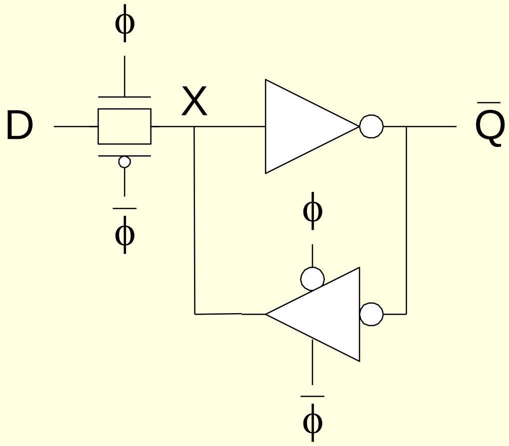
- Inverting buffer
- Both are Fast Dynamic Latches
- Pros
  - + Restoring
  - + No backdriving
  - + Fixes either
    - Output noise sensitivity
    - Or diffusion input
- Cons
  - – Inverted output





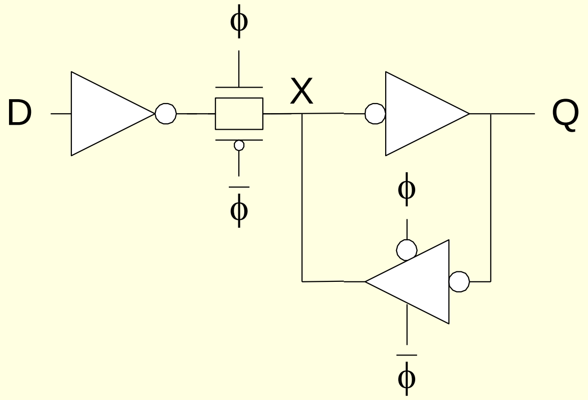
Latch Design

- Tristate feedback
- Pros
  - + Static
- Cons
  - – Backdriving risk
- Static latches are now essential because of leakage
- Reintroduced output noise sensitivity



Latch Design

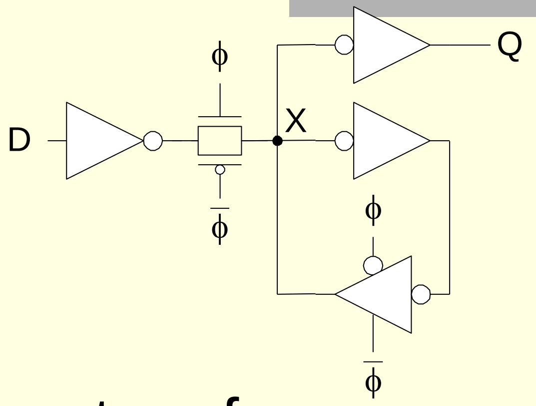
- Buffered input
  - + Fixes diffusion input
  - + Noninverting



- Reintroduced output noise sensitivity

Latch Design

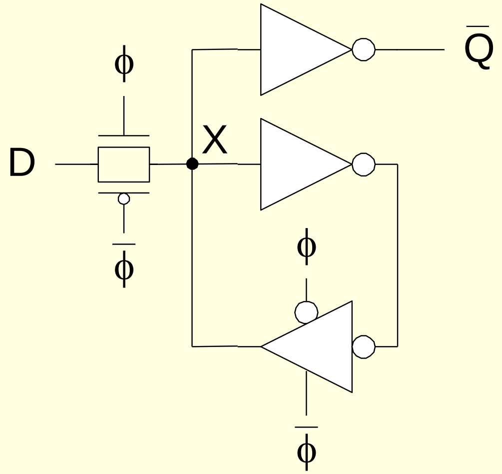
- Buffered output
  - \+ No backdriving



- Recommended for all but the most performance- or area-critical designs.
- Widely used in standard cells
  - \+ Very robust (most important)
  - \- Rather large
  - \- Rather slow (1.5 – 2 FO4 delays)
  - \- High clock loading

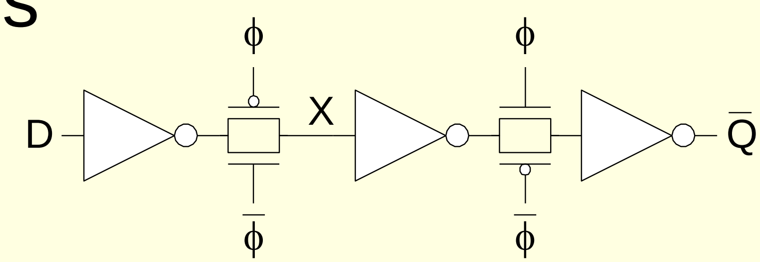
Latch Design

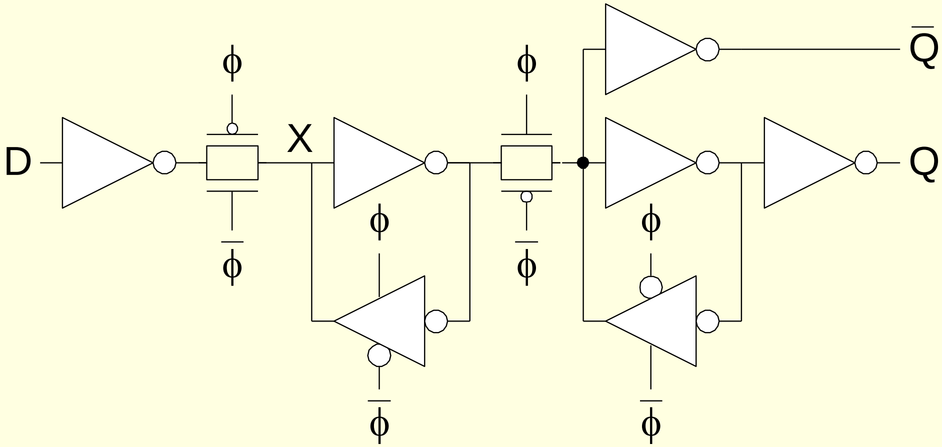
- Datapath latch
- Where input noise can be better controlled
  - + Smaller, faster
  - - unbuffered input



Flip-Flop Design

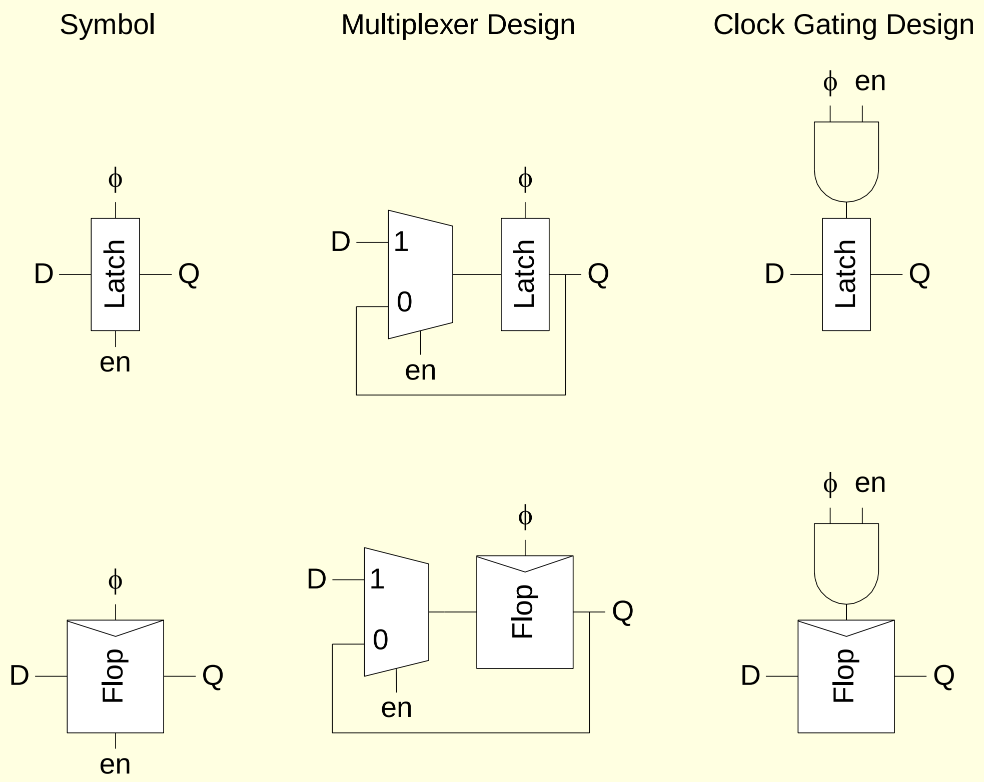
- Flip-flop is built as pair of back-to-back latches





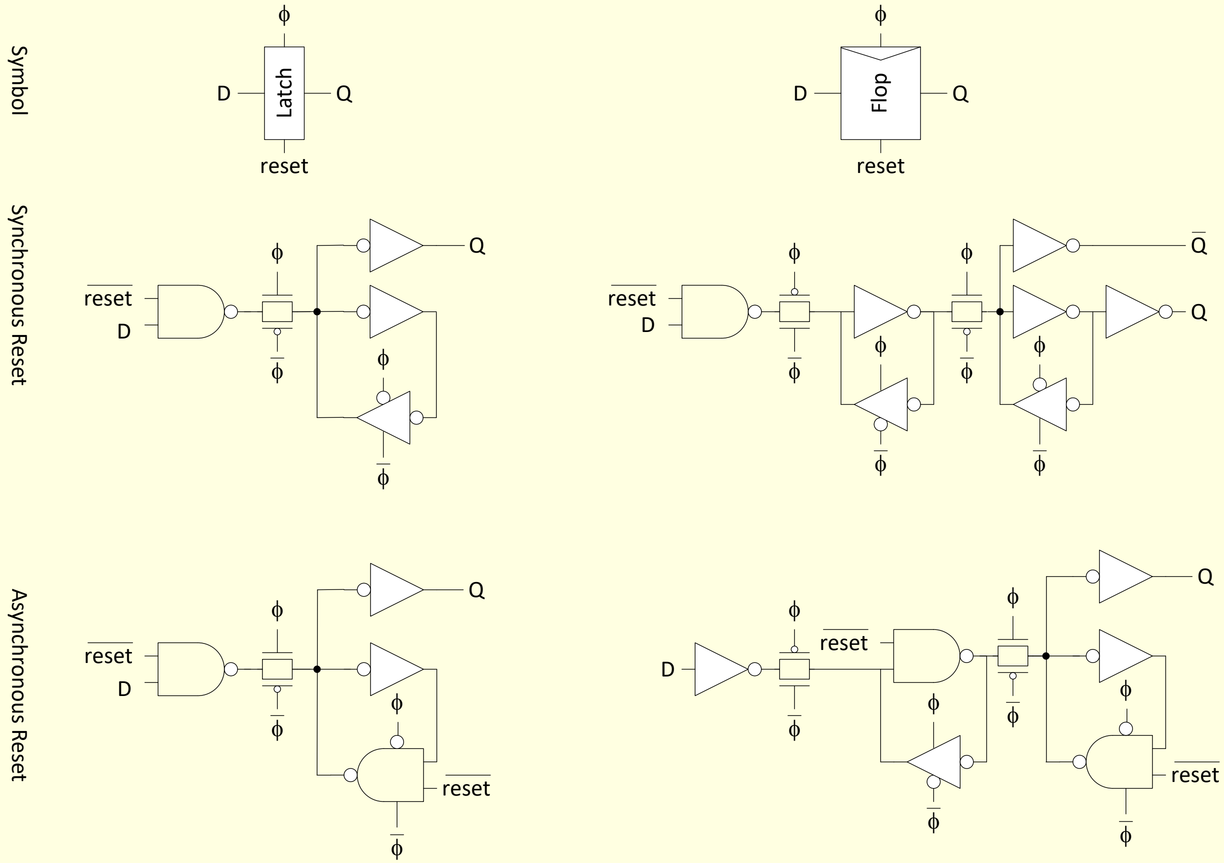
Enable

- Enable: ignore clock when en = 0
  - Mux: increase latch D-Q delay
  - Clock Gating: increase en setup time, skew



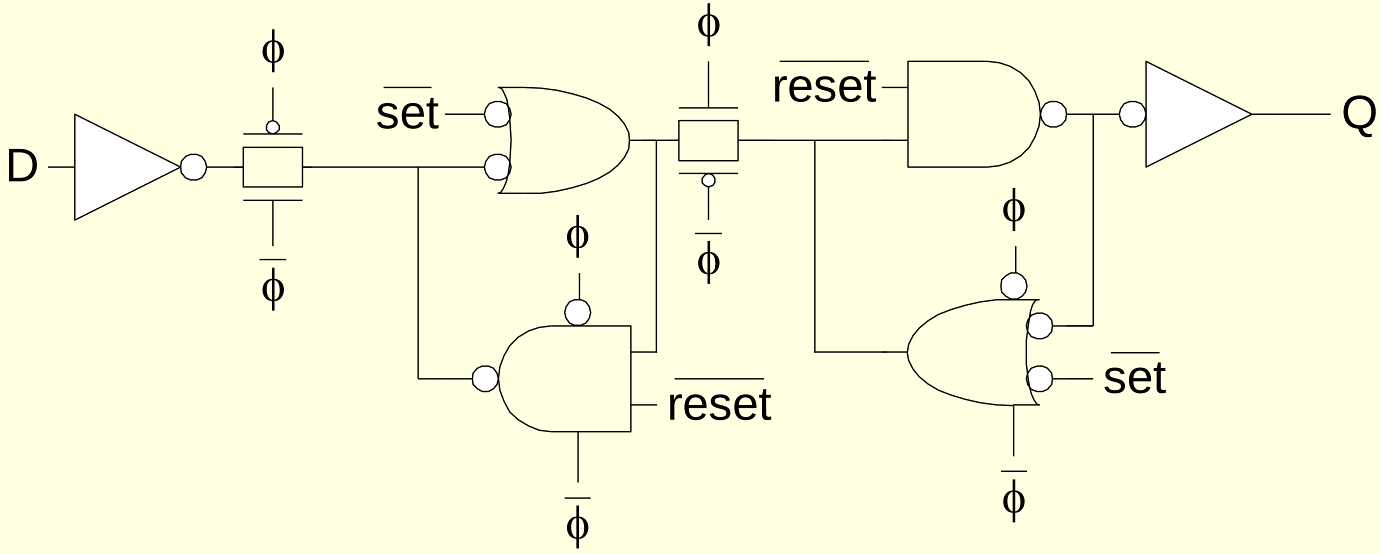
Reset

- Force output low when reset asserted
- Synchronous vs. asynchronous



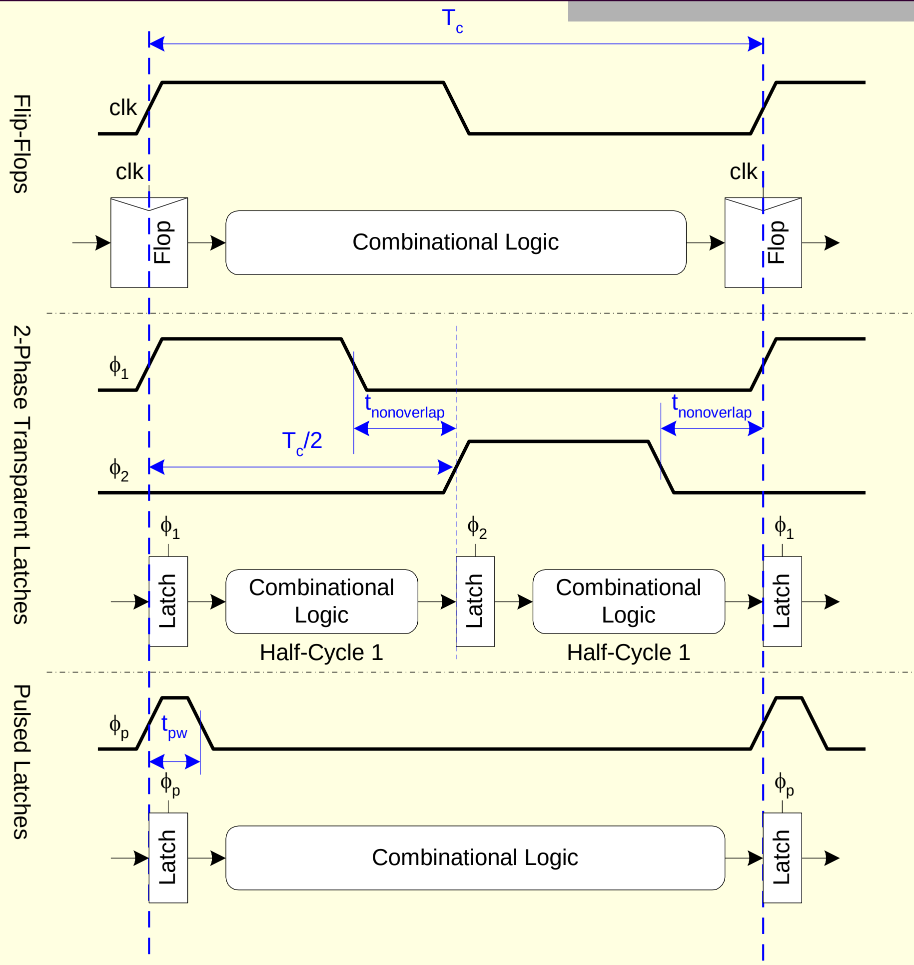
Set / Reset

- Set forces output high when enabled
- Flip-flop with asynchronous set and reset



Sequencing Methods for Static Ckt

- Flip-flops
- 2-Phase Latches
- Pulsed Latches



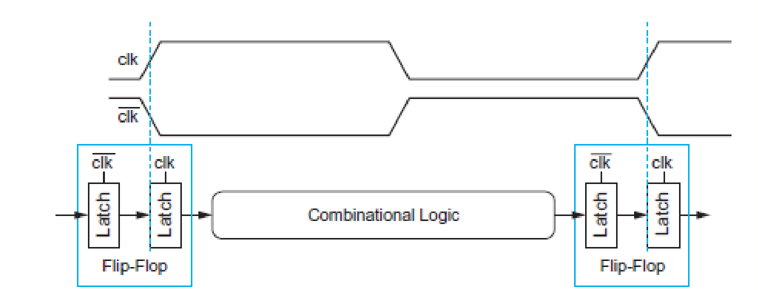
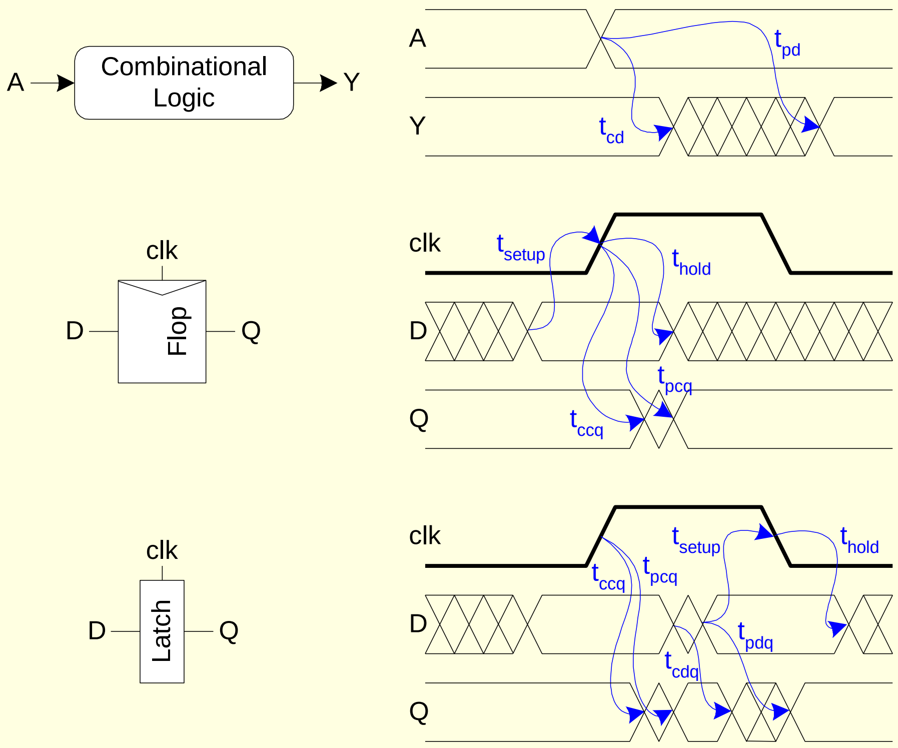


FIGURE 10.3 Flip-flop viewed as back-to-back latch pair

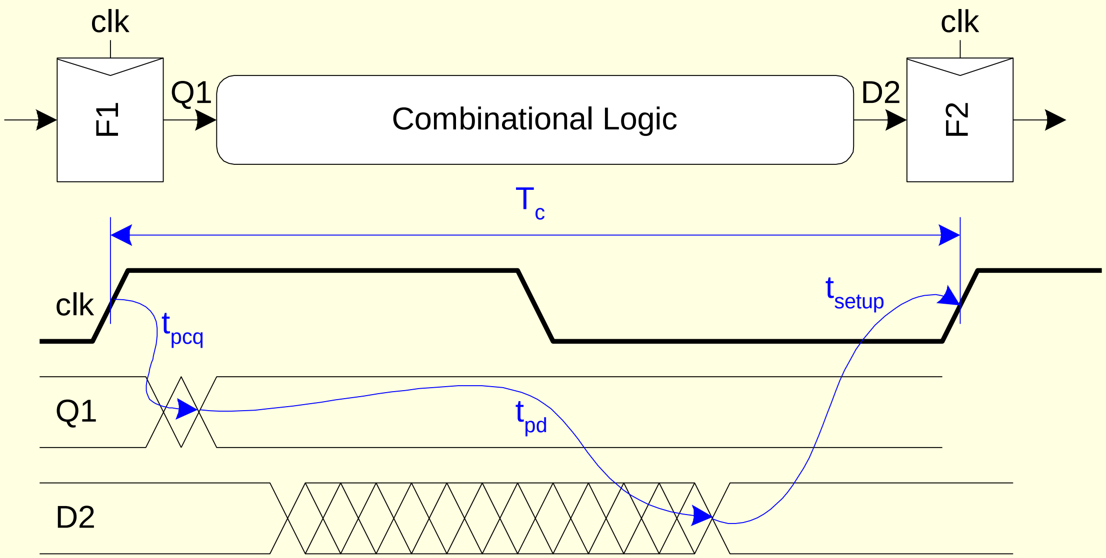
Timing Diagrams

Contamination and Propagation Delays

t_{pd}	Logic Prop. Delay
t_{cd}	Logic Cont. Delay
t_{pcq}	Latch/Flop Clk-Q Prop Delay
t_{ccq}	Latch/Flop Clk-Q Cont. Delay
t_{pdq}	Latch D-Q Prop Delay
t_{cdq}	Latch D-Q Cont. Delay
t_{setup}	Latch/Flop Setup Time
t_{hold}	Latch/Flop Hold Time



Max-Delay: Flip-Flops



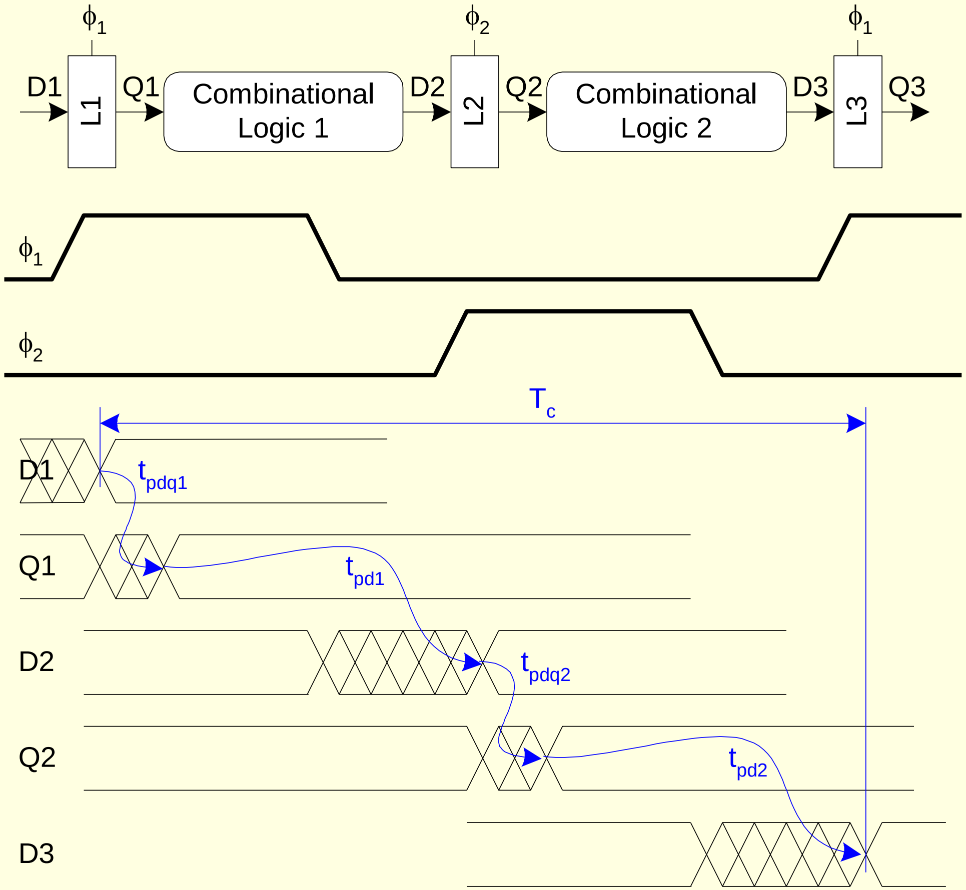
$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

$$t_{pd} \leq T_c - \underbrace{\left(t_{\text{setup}} + t_{pcq}\right)}_{\text{sequencing overhead}}$$

Max Delay: 2-Phase Latches

$$T_c \geq t_{pdq1} + t_{pd1} + t_{pdq2} + t_{pd2}$$

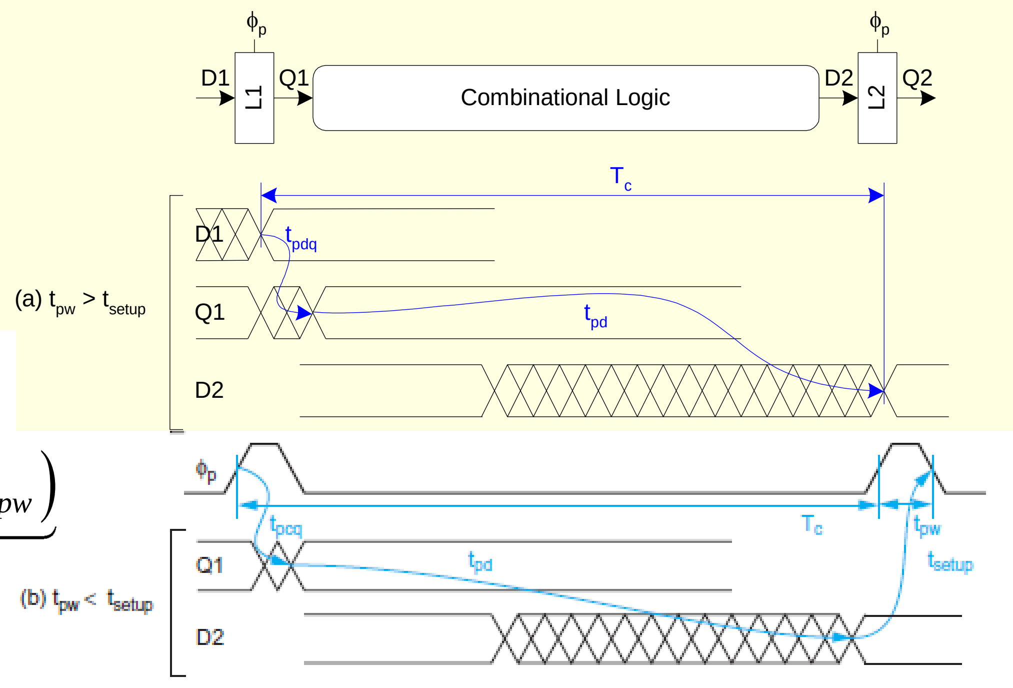
$$t_{pd} = t_{pd1} + t_{pd2} \leq T_c - \underbrace{\left(2t_{pdq}\right)}_{\text{sequencing overhead}}$$



Max Delay: Pulsed Latches

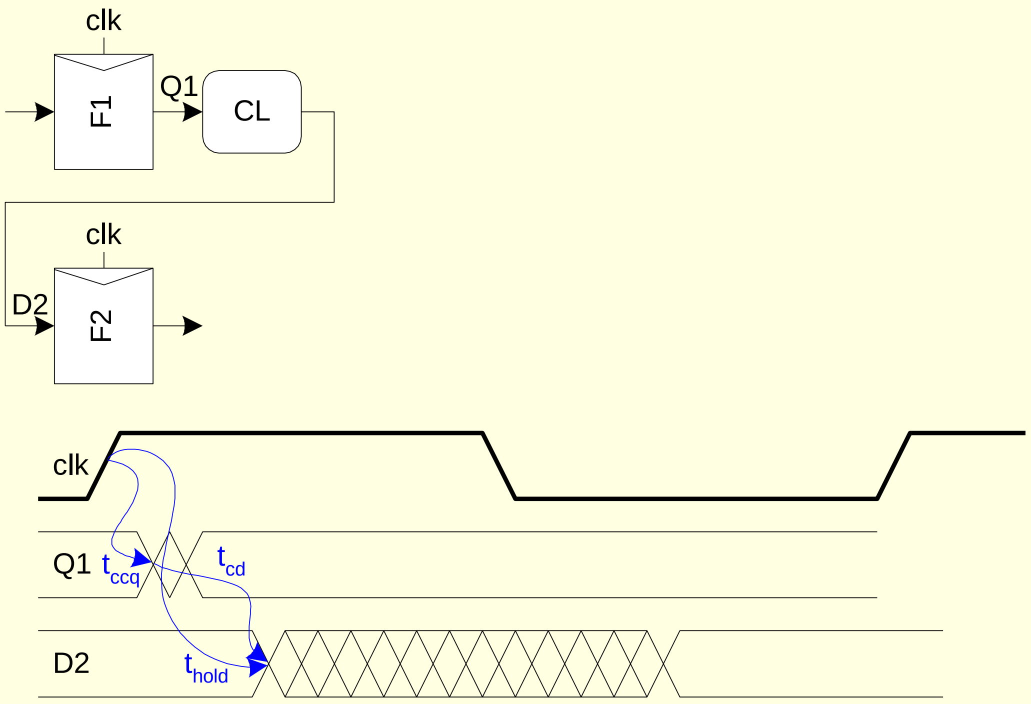
$$T_c \ge \max\left(t_{pdq}+t_{pd},\ t_{pcq}+t_{pd}+t_{setup}-t_{pw}\right)$$

$$t_{pd} \le T_c - \underbrace{\max\left(t_{pdq},\ t_{pcq}+t_{\text{setup}}-t_{pw}\right)}_{\text{sequencing overhead}}$$



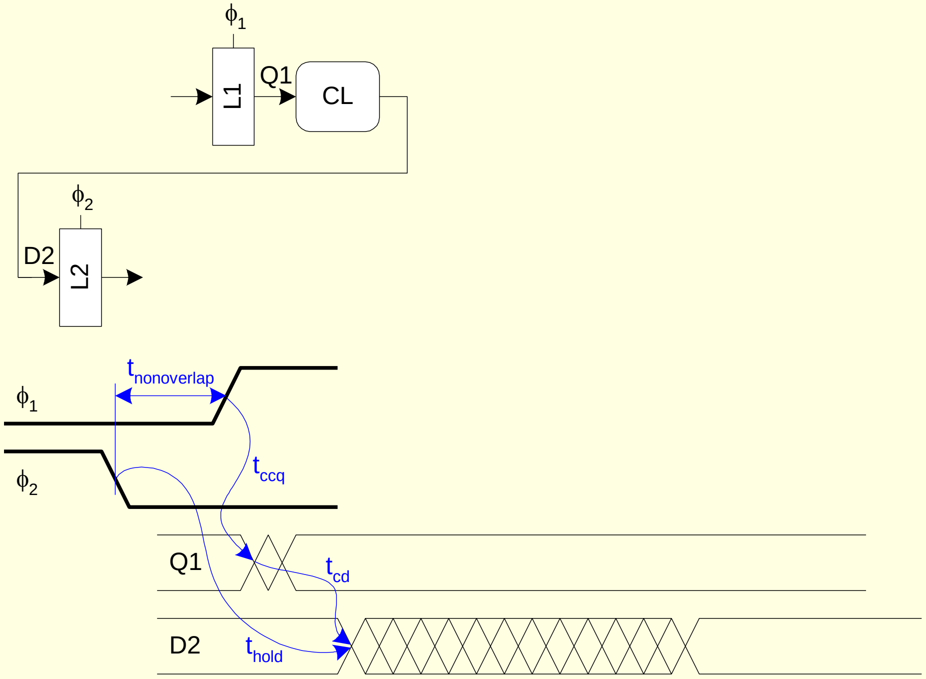
Min-Delay: Flip-Flops

$$t_{cd} \geq t_{\text{hold}} - t_{ccq}$$



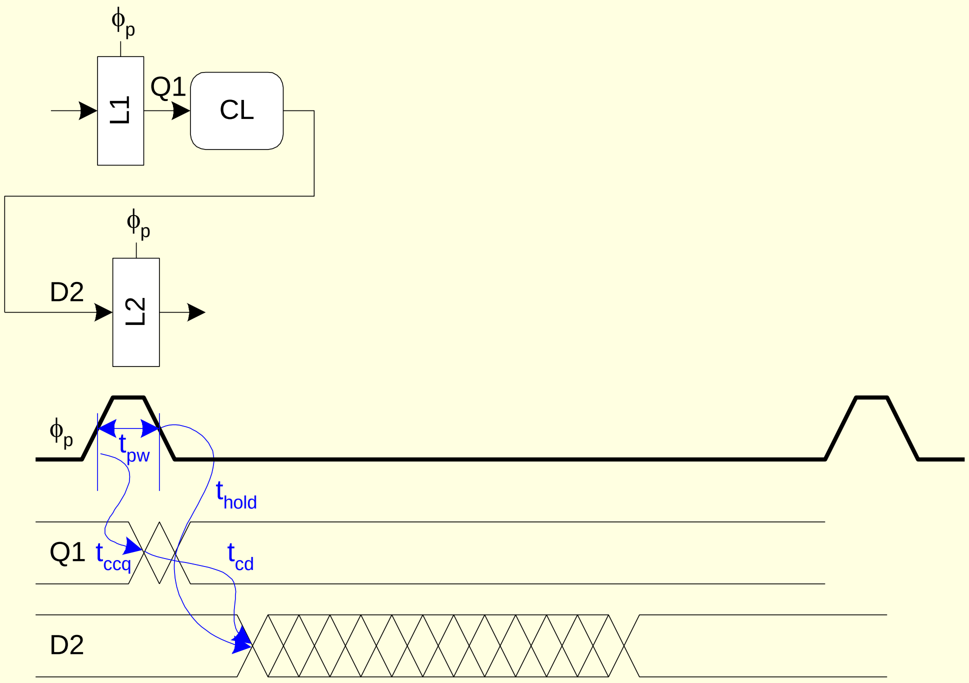
Min-Delay: 2-Phase Latches

$$t_{cd1}, t_{cd2} \geq t_{\text{hold}} - t_{ccq} - t_{\text{nonoverlap}}$$



Min-Delay: Pulsed Latches

$$t_{cd} \geq t_{\text{hold}} - t_{ccq} + t_{pw}$$



Home Work

For each of the following sequencing styles, determine the maximum logic propagation delay available within a 500 ps clock cycle. Assume there is zero clock skew and no time borrowing takes place.

a) Flip-flops

b) Two-phase transparent latches

c) Pulsed latches with 80 ps pulse width

	Setup Time	*clk*-to-*Q* Delay	*D*-to-*Q* Delay	Contamination Delay	Hold Time
Flip-Flops	65 ps	50 ps	n/a	35 ps	30 ps
Latches	25 ps	50 ps	40 ps	35 ps	30 ps